$E[X^n]$ = n^{th} moment $n = 1, 2, \ldots$

$\mu_X = E[X]$ mean, expected value

$\text{var}(X) = E[(X - \mu_X)^2]$

$E[(X-\mu_X)^n] \rightarrow n^{th}$ centralized moment

$Y = \frac{X - \mu_X}{\sigma_X} \Rightarrow E[Y] = 0$ (centralized), $\text{var}(Y) = 1$ (normalized)

Moment generating function

$M_X(t) \triangleq E[e^{tX}]$ only for t for which expectation exists

$M_X(0) = 1$ typically t: interval around '0' $M_X(t)$ exists

If X is bounded, $M_X(t)$ exists $\forall t$

$X \sim$ Bernoulli $(p) \sim (0^{1-p}, 1^{p})$

$$M_X(t) = E[e^{tX}] = (1-p) + p \cdot e^t = 1 + p(e^t - 1)$$

$X \sim$ Geometric $(p) \sim \{1, 2, \ldots\}$ $p(1-p)^{k-1}$

$$M_X(t) = \sum_{k=1}^{\infty} (1-p)^{k-1} \cdot p \cdot e^{kt} = \frac{pe^t}{1-(1-p)e^t}$$

only if $(1-p)e^t < 1$ or $t < \log \frac{1}{1-p}$

$X \sim$ poisson $(\lambda) \sim \{0, 1, \ldots\}$ $e^{-\lambda}\lambda^k / k!$

$$M_X(t) = \sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} e^{kt} = e^{\lambda(e^t - 1)} \quad \forall t$$

$X \sim$ Binomial (n, p)

$M_X(t) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} e^{tk} = (1-p+pe^t)$

$\downarrow$

$M_{Ber(p)}(t)$

If $X_1, X_2 \to$ independent:

$$M_{X_1+X_2+\dots}(t) = E[e^{t(X_1+X_2+\dots)}] = M_{X_1}(t) \cdot M_{X_2}(t) \cdots$$

$X \sim Unif[0,1]$

$$M_X(t) = E[e^{tx}] = \int_0^1 1 \cdot e^{tx} dx = \frac{e^t - 1}{t}$$

$X \sim exp(\lambda)$

$$M_X(t) = \int_0^\infty \lambda e^{-\lambda x} \cdot e^{tx} dx = \lambda/(\lambda - t) = 1 + t/\lambda + t^2/\lambda^2 + \dots$$

only if $t < \lambda$

$X \sim pdf \ \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \ \forall x$

$$M_X(t) = \int_{-\infty}^{\infty} e^{-x^2/2} e^{tx} = e^{t^2/2} = 1 + \frac{t^2}{2} + \frac{1}{2!}\frac{t^4}{2^2} + \dots$$

$$M_X(t) = E\left[1 + tx + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3!} + \dots\right]$$

$M_X(0) = 1$; $M_X'(0) = E[X]$; $M_X^{(k)}(0) = E[X^k]$

$Cov(X,Y) = E[XY] - E[X]E[Y]$

If X, Y are centralized, normalized $= S_{XY} = E[XY]$

conditional expection: Not an expectation, but a random variable

$Y \backslash X$	-1	0	1
-1	1/2	1/4	1/8
1	1/16	1/16	0

$Y/X=-1 \sim \{-1, 1\}$ with $8/9$, $1/9$

$Y/X=0 \sim \{-1, 1\}$ with $4/5$, $1/5$

$Y/X=1 \sim \{-1\}$

$Y/X = x$: distribution $\quad E[Y/X=x]$: function of x $g(x)$

g: x	$g(x)$
-1	-7/9
0	-3/5
1	-1

conditional expectation of Y given X

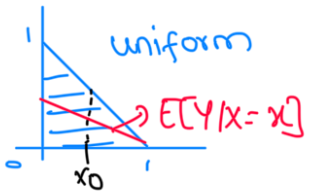
$E[Y/X] \triangleq g(X)$

$E[Y/X] \sim \{-7/9, -3/5, -1\}$ with $9/16$, $5/16$, $2/16$



$E[Y/X = x_0] = 1/2 \quad x_0 \in [0,1]$





replace x_0 with x to get $E[Y|X]$

$E[Y|X=x_0] = \frac{1-x_0}{2} \Rightarrow E[Y|X] = \frac{1-X}{2} \in [0, \frac{1}{2}]$

$X \sim$ pdf $2(1-x)$

$Y = g(X) = \frac{1-X}{2}$ $\quad f_Y(y) = \frac{f_X(1-2y)}{|-1/2|} = 8y$

Properties:

1) Tower property: $E[E[Y|X]] = E[Y]$

$X_1, X_2, \ldots X_n \sim$ iid Bernoulli(p)

$Y = \min\{k : X_k = 1\} \sim$ Geometric (p)

$E[Y|X_1 = 1] = 1$ $\qquad E[Y|X_1 = 0] = 1 + E[Y]$

$$E[Y|X_1] \sim \begin{cases} p & 1-p \\ 1 & 1+E[Y] \end{cases}$$

$E[Y] = E[E[Y|X_1]] = E[Y|X_1=0]\,P[X_1=0] + E[Y|X_1=1]\,P[X_1=1]$

($1+E[Y]$, $1-p$, 1, p)

$\Rightarrow E[Y] = 1/p$

$Z = \min\{k : X_{k-1} = 0, X_k = 1\}$

$E[Z] = ?$

2) Prediction:

X, Y joint distribution

Observe X, predict Y using only X & joint distribution

$\Rightarrow$ I'm looking for a function of X $g(X)$ such that $g(X)$ is "predicts" Y well

mean square error (MSE): $E[\underbrace{(Y-g(X))^2}_{\text{error}}]$

$\arg\min E[(Y-g(X))^2] = E[Y|X]$